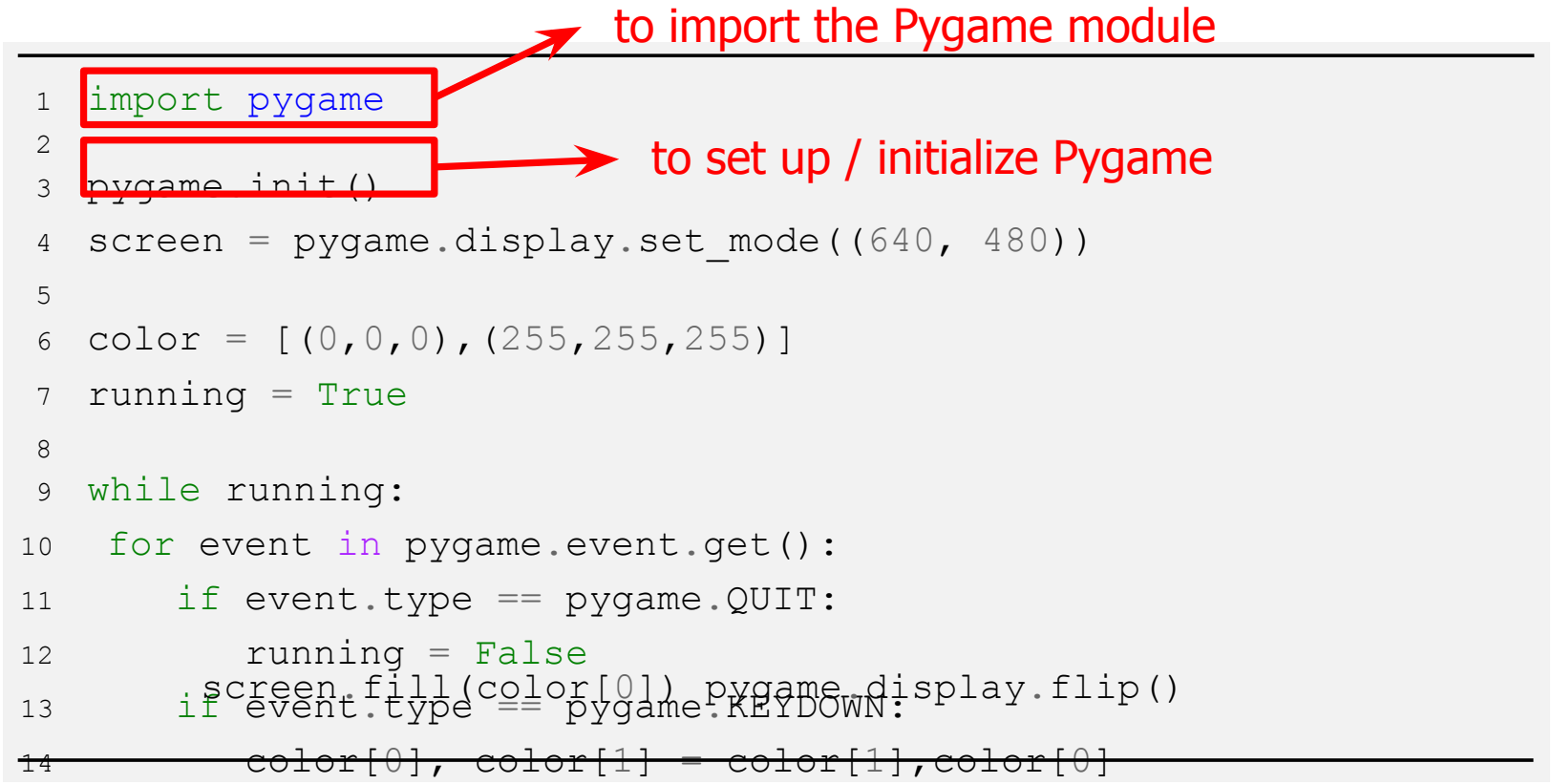
Pygame!

A simple Pygame example

```
1  import pygame
2
3  pygame.init()
4  screen = pygame.display.set_mode((640, 480))
5
6  color = [(0,0,0),(255,255,255)]
7  running = True
8
9  while running:
10  for event in pygame.event.get():
11      if event.type == pygame.QUIT:
12          running = False
13      if event.type == pygame.KEYDOWN:
14          color[0], color[1] = color[1],color[0]
15  screen.fill(color[0]) pygame.display.flip()
16
17
```

01-simple_pygame.py

What each element does: **Import and Initialize**

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01-simple_pygame.py

A simple Pygame example

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16
17
```

01-simple_pygame.py

What each element does: **Setting Window & Screen**

```
screen = pygame.display.set_mode((640, 480))
```

initializes a **window** with dimensions 640 × 480 and returns the **screen object.**

Everything to be displayed needs to be drawn on the `screen`.

A simple Pygame example

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6  color = [(0,0,0),(255,255,255)]
7  running = True
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11      if event.type == pygame.QUIT:
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14          color[0], color[1] = color[1],color[0]
15  screen.fill(color[0]) pygame.display.flip()
16
17
```

01-simple_pygame.py

What each element does: **The Main Loop**

```
1  running = True while
2  running:
3
4      for event in pygame.event.get():
5          if event.type == pygame.QUIT:
6              running = False
7          if event.type == pygame.KEYDOWN:
8
9              color[0], color[1] = color[1],color[0]
10
```

```
    screen.fill(color[0])
    pygame.display.flip()
```

is the **main loop** of the game.

- listen to events (closing game, pressing keys) → respond
- proceed the game
- draw on the screen
- stop when done

Drawing

Drawing on the screen

Filling the screen with a color:

```
1 blue = (0,0,255)
2 screen.fill(blue)
3 pygame.display.flip()
```

After all drawing is done, call display.flip() to **update** the display.

Use pygame.draw to draw geometric shapes. A circle:

```
1 red = (255,0,0)
2 # position (320,240), radius = 50
3 pygame.draw.circle(screen, red, (320,240), 50)
```

Drawing on the screen

Geometric shapes available for pygame.draw:

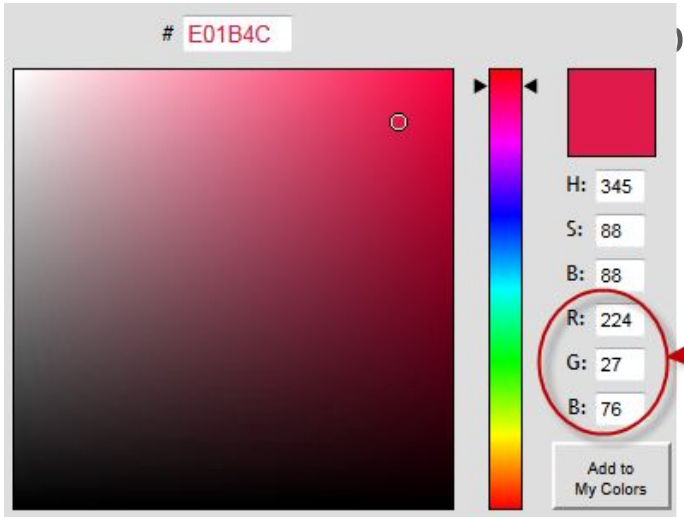
- circle(Surface, color, pos, radius, width=0)
- polygon(Surface, color, pointlist, width=0)
- line(Surface, color, start, end, width= 1)
- rect(Surface, color, Rect, width=0)
- ellipse(Surface, color, Rect, width=0)

```
1 red = (255,0,0)
2 pygame.draw.line(screen, red, (10,50),(30,50),10)
```

Drawing on the screen - **Colors**

Defining a color

```
gray = (200,200,200)
#(red, green, blue)
```

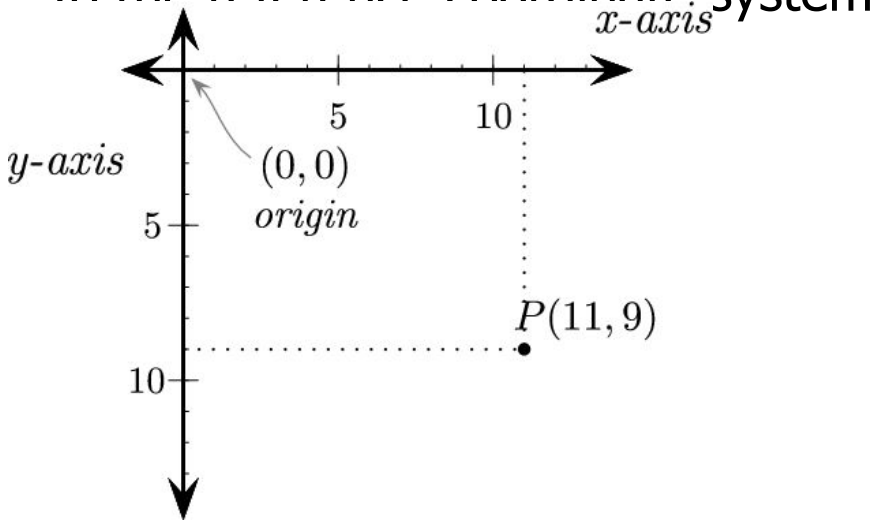


Drawing on the screen - **Positions**

Defining a position:

```
P = (11,9)
#(x-axis, y-axis)
```

To the reference coordinate system



Drawing on the screen - **Rects**

```
pygame.Rect(left, top, width, height)
```

to create a Rect (i.e., rectangle)

```
1  box = pygame.Rect(10, 10, 100, 40)
2  pygame.draw.rect(screen, blue, box)
3  #draws at (10,10) rectangle of width 100, height 40
```

A full drawing example

```
1  import pygame
2
3  pygame.init()
4  screen = pygame.display.set_mode((640, 480))
5
6  white = (255,255,255)
7  blue = (0,0,255)
8
9  running = True
10 while running:
11  for event in pygame.event.get():
12      if event.type == pygame.QUIT:
13          running = False
14      screen.fill(white)
15      pygame.draw.circle(screen, blue, (320,240), 100)
16      # position (320,240), radius = 100
17      pygame.display.flip()
18
```

04-drawing.py.

User Input

Events

get all events in Pygame's event queue

```
pygame.event.get()
```

usually used as

```
for event in pygame.event.get():
    if event.type == YourEvent:
        react_to_your_event()
```

Event types

Some of the **most important** event types are:

```
pygame.QUIT
pygame.KEYDOWN
pygame.KEYUP
pygame.USEREVENT
```

Events

React to KEYDOWN event:

```
while running:
    for event in pygame.event.get():
        if event.type == KEYDOWN:
            react_to_key()
```

Which key? → if event type is KEYDOWN or KEYUP event has attribute **key**.

```
for event in pygame.event.get(): if
    event.type == KEYDOWN:
        if event.key == K_ESCAPE: running
            = False
```

Pygame Keys

Some of the most important keys are:

```
K_RETURN K_SPACE
K_ESCAPE
K_UP, K_DOWN, K_LEFT, K_RIGHT
K_a, K_b, ...
K_0, K_1, ...
```

Full list of keys: http://www.pygame.org/docs/ref/key.html

Getting continuous input

`KEYDOWN` is a unique event.

```
key = pygame.key.get_pressed()
```

to get keys currently pressed.

```
if key[pygame.K_UP]: move_up()
```

to check and react on a specific key.

A user input example

```
color = [0,0,0]

while running:
 for event in pygame.event.get():
    if event.type == pygame.QUIT:
       running = False
    if event.type == KEYDOWN and event.key == K_SPACE:
       color = [0,0,0]
     keys = pygame.key.get_pressed()
     if keys[K_UP]:
         color = [(rgb+1)%256 for rgb in color]

     screen.fill(color) pygame.display.flip()
```

05-user_input.py

Pygame’s Clock

Limiting the frames per second

```
clock = pygame.time.Clock()
```

to initialize the clock.

In your main loop call

```
clock.tick(60) #limit to 60 fps
```

Clock example

```
clock = pygame.time.Clock()

while running:
  for event in pygame.event.get():
     if event.type == pygame.QUIT:
        running = False
     if event.type == KEYDOWN and event.key == K_SPACE:
        color = [0,0,0]
     keys = pygame.key.get_pressed()
     if keys[K_UP]:
         color = [(rgb+1)%256 for rgb in color]

     screen.fill(color)
     pygame.display.flip()
     clock.tick(60)

```

06-clock.py

Game Events

Sprites

Pygame's **sprite class** gives convenient options for handling interactive graphical objects.

If you want to draw **and** move (or manipulate) and object, make it a sprite.

Sprites

can be grouped together
are easily drawn to a surface (even as a group!) have
an update method that can be modified have collision
detection

Basic Sprite

```
1  class SpriteExample(pygame.sprite.Sprite):
2
3      def __init__(self): pygame.sprite.Sprite.__init
4          (self)
5
6          self.image = #image
7          self.rect = #rect
8
9      def update(self): pass
10
```

Sprite from a local image

```
1  class SpriteExample(pygame.sprite.Sprite):
2
3      def __init__(self): pygame.sprite.Sprite.__init
4          (self)
5
6          self.image = pygame.image.load('local_img.png') self.rect =
7          self.image.get_rect()
8          self.rect.topleft = (80,120)
9
10     def update(self): pass
11
```

Self-drawn sprites: Rectangle

```
1  class Rectangle(pygame.sprite.Sprite):
2
3      def __init__(self): pygame.sprite.Sprite.__init
4          (self)
5          self.image = pygame.Surface([200, 50])
6          self.image.fill(blue)
7          self.rect = self.image.get_rect() self.rect.top,
8          self.rect.left = 100, 100
9
10     def update(self): pass
11
```

Sprites: **A reminder about classes**

```
class Rectangle(pygame.sprite.Sprite):

 def __init__(self, color):
        pygame.sprite.Sprite.__init__(self)
        self.image = pygame.Surface([200, 50])
        self.image.fill(color)
            self.rect = self.image.get_rect()
            self.rect.top, self.rect.left = 100, 100

    def update(self):
            pass
```

```
white, blue = (255,255,255), (0,0,255)

white_rect = Rectangle(white)
blue_rect = Rectangle(blue)
```

Sprite groups

```
new_sprite_group = pygame.sprite.Group(sprite1)
```

to create a new sprite group containing sprite sprite1.

```
new_sprite_group.add(sprite2)
```

to add another sprite later.

Group updating and drawing:

```
#inside main loop:

new_sprite_group.update() #game events
new_sprite_group.draw()   #drawing
```

Framework for working with sprite groups

```
1  class NewSprite(pygame.sp...
2      #defining a new sprite
3
4  newsprite = NewSprite()
5  sprites = pygame.sprite.Group()
6  sprites.add(newsprite)
7
8  while running:
9      for event in …
10        sprites.update() #game events
11        sprites.draw()
12        pygame.display.flip()
13
```

Sprite groups: Working example

```
1  class Circle(pygame.sprite.Sprite):
2   def __init__(self):
3      pygame.sprite.Sprite.__init__(self)
4      self.image = pygame.Surface([100, 100])
5      pygame.draw.circle(self.image, blue, (50, 50), 50)
6      self.rect = self.image.get_rect()
7      self.rect.center = [320,240]
8
9  def draw(sprites):
10  screen.fill(white)
11  sprites.draw(screen)
12  pygame.display.flip()
13
14 circle = Circle()
15 sprites = pygame.sprite.Group(circle)
16
17 while running:
```

07-sprite-circle.py

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